

Technical Documentation

NHS England A&E 4-Hour Performance Dashboard — the full process behind the Executive Overview and Trust Drill-Down pages, including every data problem found along the way.

40

monthly source files

214

organisations profiled

3

trust mergers confirmed

6

data anomalies caught

01

The Problem

A Regional Director of Urgent & Emergency Care needs to decide which trusts in their region need support to reach the national 4-hour A&E target. A regional average close to target can hide a wide spread — some trusts comfortably ahead, others badly behind — and without knowing which trusts are pulling the average down, support gets spread evenly instead of where it's needed.

The dashboard ranks trusts by **patients short of target**, not raw percentage — a 5-point miss at a 30,000-attendance trust represents more unmet need than a 10-point miss at a 3,000-attendance trust, and a percentage-only ranking gets that backwards.

02

Data & Scope

Source: [NHS England A&E Attendances and Waiting Times](#) — 40 monthly CSVs, April 2023 to

July 2026, no month column in the raw files (derived from each file's name during import).

Grain: one row = one organisation, one month. **Scope:** only organisations with nonzero Type 1 (major A&E) attendance in at least one month are included — a filter applied at Org Code level after combining all files, not per-file. This is a stated, data-inferred assumption, not verified against an official ODS provider-type list.

Target source: 76% (FY2023/24), 78% (FY2024/25 & FY2025/26, the latter independently verified against primary NHS planning guidance), 82% (FY2026/27 onward) — all sourced from NHS England's own fiscal-year planning documents, not secondary reporting.

03

Data Quality Investigation

Every anomaly below was traced to a specific cause and either fixed or documented as an explained limitation — none were assumed or silently dropped.

Hidden national-aggregate row **critical**

One row carried a national monthly total (1,466,119 Type 1 attendances) disguised as an ordinary organisation row, inflating every descriptive statistic. Found via Excel descriptive stats — a single-trust maximum that size is implausible. Removed; post-removal maximum dropped to a realistic 36,434.

Inconsistent footer rows

Every file ends with a "Total" row, but casing/spacing varies across five distinct forms (TOTAL, Total, TOTA1, "Total ", "TOTAL ") — roughly splitting before/after mid-2024, suggesting an NHS export format change. Fixed with a trim+uppercase filter rather than an exact-match list, so it survives any further casing variant.

September 2024 schema drift

One file carries six extra blank trailing columns and a stray header named "a." Verified empty in every row before being safely dropped — combine logic keyed by column name, not position, so the extra columns never shifted anything else.

RK9 — false 100% performance, April & May 2023

Zero recorded breaches against a normal ~8,000 monthly attendance volume, two consecutive months. Confirmed as a genuine source submission error via both a Power BI table and an independent raw-CSV check. Documented and excluded from that trust's trend narrative — not corrected in the source.

NDA57 — impossible admission value

Shows 1 admission against 0 attendances in April 2023. Traced to Haslemere Minor Injuries Unit, a Type 3 facility with no emergency admission pathway by design. Out of scope (zero Type 1 activity throughout its history) so it does not affect any measure — logged as a documented, uncorrected source anomaly.

04

Organisation Register

Of 214 distinct organisations, 34 had incomplete month coverage. Each was investigated individually by name, not assumed from the gap pattern alone.

Predecessor	Successor	Date	Verification
RAP North Middlesex	RAL Royal Free London	1 Jan 2025	Step-jump confirmed in successor's chart
RVJ North Bristol	RA7 Bristol NHS FT	1 Jul 2026	Step-jump confirmed in successor's chart
RVY Southport & Ormskirk	RBN Mersey & West Lancs	1 Jul 2023	Exit window matches dissolution date exactly

The remaining ~31 discontinuous codes were checked by organisation name and confirmed to be GP out-of-hours services, walk-in centres, and urgent treatment centres — minor-care providers outside the dashboard's Type 1 scope, not mergers or errors.

Three predecessor→successor pairings were initially hypothesised purely from date proximity and disproven once real organisation names were checked — a standing reminder that timing coincidence is not evidence.

05

Data Model

- **Fact table** — one row per in-scope organisation per month, all attendance/breach/admission measures.
- **Org Code dimension** — Org Code only, deliberately with no name attached. Organisation names are not stable (two trusts gained "Teaching" status mid-series), so names stay in the fact table as historical, point-in-time attributes rather than being forced into a one-name-per-code lookup.
- **dim_calendar** — independently generated daily calendar (not derived from which months exist in the data), April 2023–July 2026, with fiscal-year columns throughout (NHS runs April–March). Marked as the model's official Date table.
- Relationships: single-direction, many-to-one, dimension → fact. No ambiguous paths.

06

Measure Catalog

Every measure was validated by hand against the raw source CSVs before being trusted — no measure went into the dashboard unverified.

Measure	What it does	Validated
Type 1 4-Hour Performance %	Core headline rate	RA7 Apr-23 = 66.25% match
Dynamic Target %	Applicable NHS target by fiscal year	4 values, each independently sourced
Variance	Performance – Target, in points	RA7 Apr-23 = –9.8pp match
Gap in Patients	Patients short of target, by volume	RRK Jul-26 = 7,314 match
Type 1 Performance YoY	Change vs. same month last year	RA7 Apr-24 vs Apr-23 = –1.7pp match
Type 1 12-Month Rolling %	Trailing 12-month rate, totals-weighted	RA7 Mar-24 = 63.3% match

Admission Conversion Rate %	Share of attendances resulting in admission	RA7 Apr-24 = 33.9% match
Type 1 Status	Distinguishes no-data from zero-activity months	Logic-tested on known merger cases

Gap in Patients replaced an earlier Variance × Volume approach, rejected for producing a unit with no real meaning. The chosen formula returns an actual patient count and sorts correctly without extra handling for over-performing trusts.

07

Dashboard Architecture

Two pages, each answering one question — deliberately not combined into one dense view, given the audience (a Regional Director, reviewing quarterly) is a strategic decision-maker, not a day-to-day analyst.

Page 1 — Executive Overview

"Where do I focus right now?" Four KPI cards, a trust table ranked by Gap in Patients, a month/region filter defaulting to the latest complete month, and a plain-language insight sentence stating the regional picture.

Page 2 — Trust Drill-Down

"Is this a new problem or a persistent one?" Reached by drilling into any trust row. Full 40-month trend against the target line, a patient-gap column chart, and a clickable period card showing exact figures for whichever month is selected.

08

Key Findings

- 1. Not every low performer has the same story.** One trust has run flat, underperforming for the full 40 months — its target moved, not its performance. Another trust genuinely improved for two years, then declined sharply. Treating both as "below target" would miss that they need different interventions entirely.

miss that they need different interventions entirely.

2. **Percentage rankings and volume-weighted rankings disagree.** The trust with the single worst percentage miss is rarely the trust with the most patients affected — ranking by raw percentage alone would have sent support to the wrong places.
3. **The best performer by percentage isn't the best performer by scale.** The dashboard highlights the trust handling the largest patient volume while still beating target, not the smallest trust with the highest percentage.
4. **Winter pressure is real but not uniform.** A seasonal dip (roughly November–January) appears in every year of data, but its depth and shape vary meaningfully year to year — flagged directly in the dashboard so a month-to-month comparison isn't misread as a new problem.

09

Known Limitations

- One merger (RAP → RAL) is validated at chart level only, not yet with a row-level raw-CSV spot check like the other two mergers received.
- "How does this month compare to last winter?" isn't answerable at a glance — it requires scrolling the trend chart manually.
- Row-Level Security and mobile layout are explicitly out of scope — single fictional audience, portfolio deployment via GitHub rather than Power BI Service.
- 14 raw columns (secondary attendance/admission categories, long-wait indicators) are present in the model but unused by any current measure — kept rather than removed, to avoid a one-way Power Query change for a possible future extension.